Kiriş-Kolon Hesabı

Kiriş – Kolon

- **Eğilme Momenti ve Eksenel Kuvvet Etkisindeki Çift ve Tek Simetri Eksenli Elemanlar (11.1)**
  - **Eğilme ve Basınç Etkisindeki Çift ve Tek Simetri Eksenli Elemanlar**
  - **Eğilme ve Çekme Etkisindeki Çift ve Tek Simetri Eksenli Elemanlar**
  - **Tek Eksenli Eğilme ve Basınç Etkisindeki Çift Simetri Eksenli Kompakt Enkesitli Hadde Elemanlar**
- **Eğilme Momenti ve Eksenel Kuvvet Etkisindeki Diğer Elemanlar (11.2)**
- **Burulma Etkisindeki Elemanlar ve Burulma, Eğilme, Kesme ve/veya Eksenel Kuvvetin Ortak Etkisindeki Elemanlar (11.3)**
  - **Burulma Etkisindeki Boru ve Kutu Enkesitli Elemanlar**
  - **Burulma, Kesme Kuvveti, Eğilme ve Eksenel Kuvvetin Ortak Etkisindeki Boru ve Kutu Enkesitli Elemanlar**
  - **Burulma ve Bileşik Gerilme Etkisindeki Boru ve Kutu Enkesitler Dışındaki Diğer Tüm Elemanlar**

Eğilme Momenti ve Eksenel Kuvvet Etkisindeki Çift ve Tek Simetri Eksenli Elemanlar

For $\frac{P_r}{P_c} \geq 0.2$,

$$\frac{P_r}{P_c} + \frac{8}{9}\left(\frac{M_{rx}}{M_{cx}} + \frac{M_{ry}}{M_{cy}}\right) \leq 1.0$$

For $\frac{P_r}{P_c} < 0.2$,

$$\frac{P_r}{2P_c} + \left(\frac{M_{rx}}{M_{cx}} + \frac{M_{ry}}{M_{cy}}\right) \leq 1.0$$

⇨

LRFD Interaction Equations

For $\frac{P_u}{\phi_c P_n} \geq 0.2$,

$$\frac{P_u}{\phi_c P_n} + \frac{8}{9}\left(\frac{M_{ux}}{\phi_b M_{nx}} + \frac{M_{uy}}{\phi_b M_{ny}}\right) \leq 1.0$$

For $\frac{P_u}{\phi_c P_n} < 0.2$,

$$\frac{P_u}{2\phi_c P_n} + \left(\frac{M_{ux}}{\phi_b M_{nx}} + \frac{M_{uy}}{\phi_b M_{ny}}\right) \leq 1.0$$

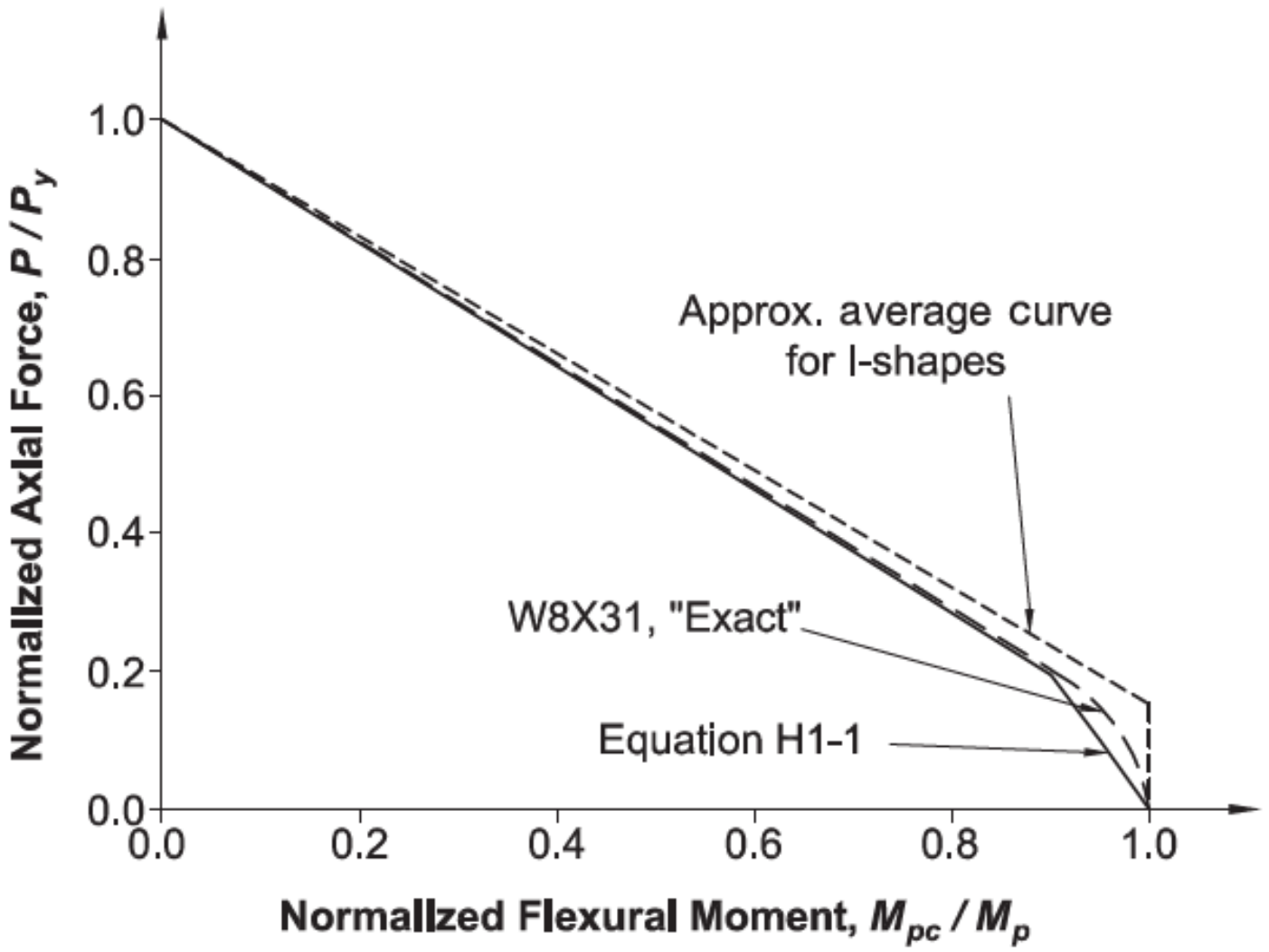


Fig. C-H1.1. Stub-column interaction curves: plastic moment versus axial force for wide-flange shapes, major-axis flexure

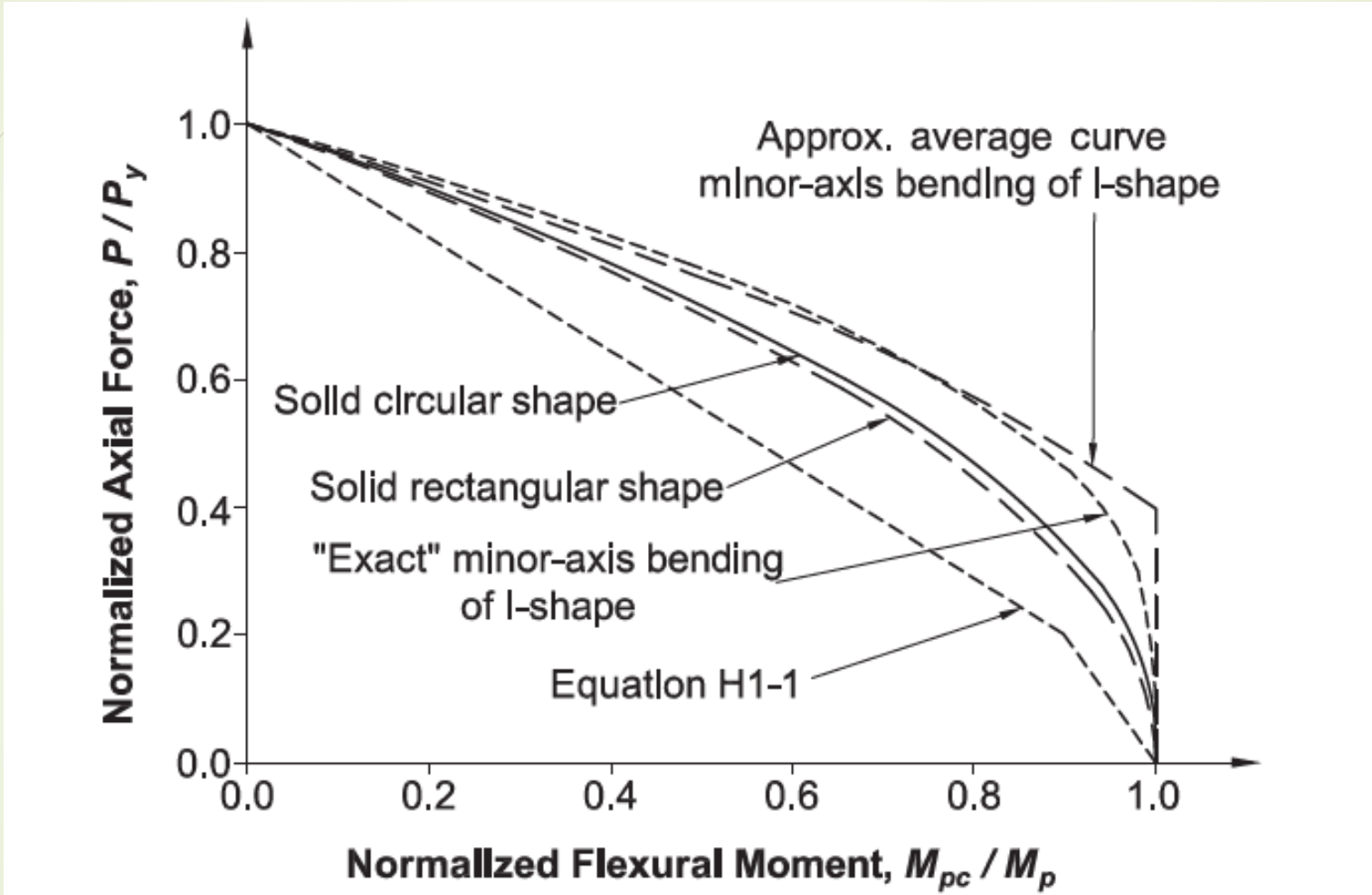
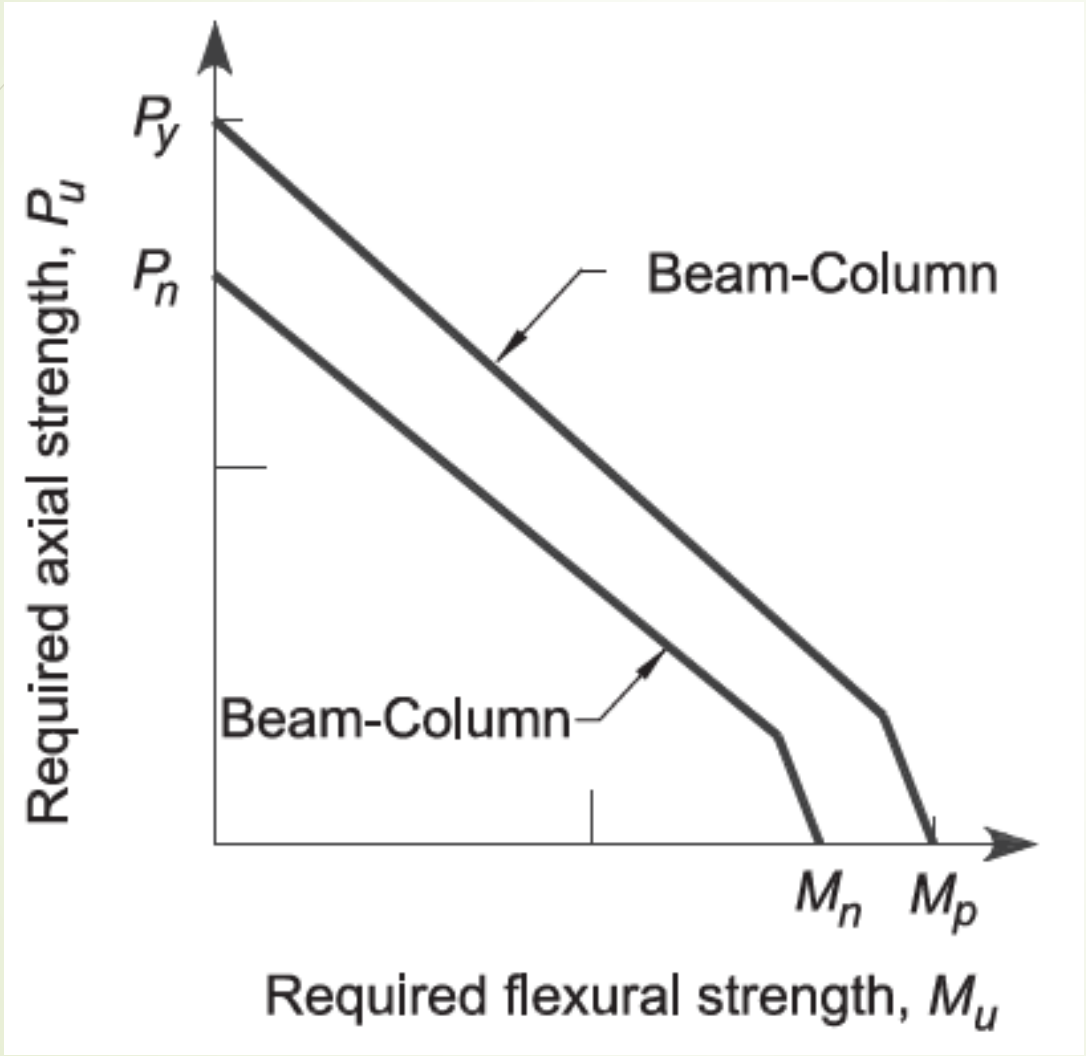
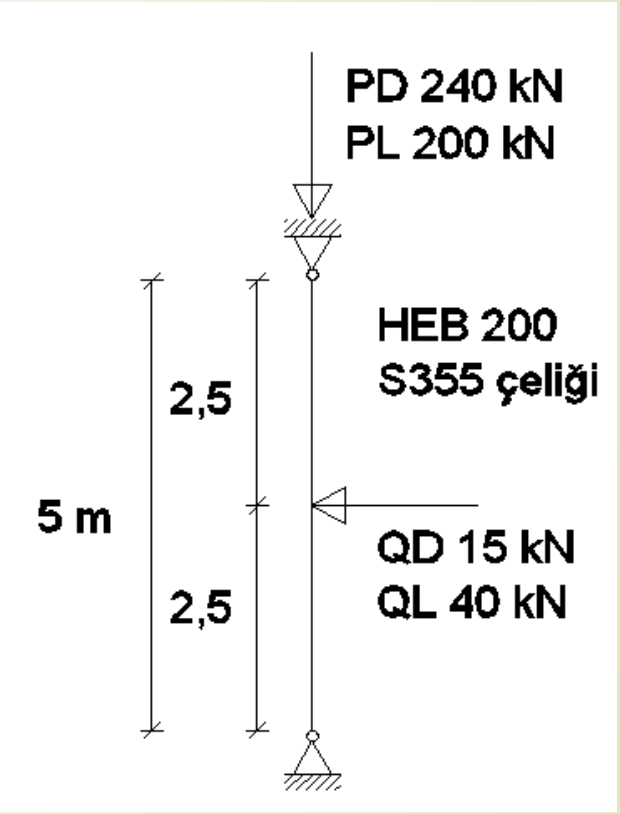


Fig. C-H1.2. Stub-column interaction curves: plastic moment versus axial force for solid round and rectangular sections and for wide-flange shapes, minor-axis flexure.


Required axial strength, P_u
P_y
P_n
Beam-Column
Beam-Column
M_n
M_p
Required flexural strength, M_u

Örnek

Şekilde görülen kiriş kolon örneğine gelen yükler ve sınır koşulları verilmiştir. Ortadaki yatay yük profil güçlü ekseni doğrultusunda etkidiği düşünülürse şartnameye göre profilin yükler altında güvenli olduğunu kontrol ediniz.



Basınç çubukları konusundan HEB200 profilinin 500 cm uzunluğundaki durumu için taşıma kapasitesi

- *Güçlü eksen doğrultusunda 0,9*2141=1927 kN*
- *Zayıf eksen doğrultusunda 0,9*1333=1200 kN olarak bulunur.*

Bu durumda profilin taşıma kapasitesi 1200 kN olarak bulunmuş olur.

Kolon temiz açıklığı (desteklenmemiş uzunluk) 500 cm olduğundan ve dersin başında anlatılan standart tablolar Cb değeri olarak bakıldığında Cb=1,32 olarak bulunmuş olur.

Eğilme Momenti hesabından HEB200 için Moment kapasitesi

- Akma durumu dikkate alındığında 0,9*22809=20527 kNcm
- Yanal burulmalı burkulma durumu için 0,9*25739= 23165 kNcm bulunur.

Bu durumda profilin güçlü eksen doğrultusundaki eğilme kapasitesi

Mcx=205,27 kNm olarak bulunmuş olur.

Zayıf eksen doğrultusunda moment oluşmadığı için kapasite hesaplamaya gerek yoktur. Fakat hesap edilseydi Mcy=9770 kNcm olarak bulunurdu.

P=240*1,2+200*1,6=608 kN

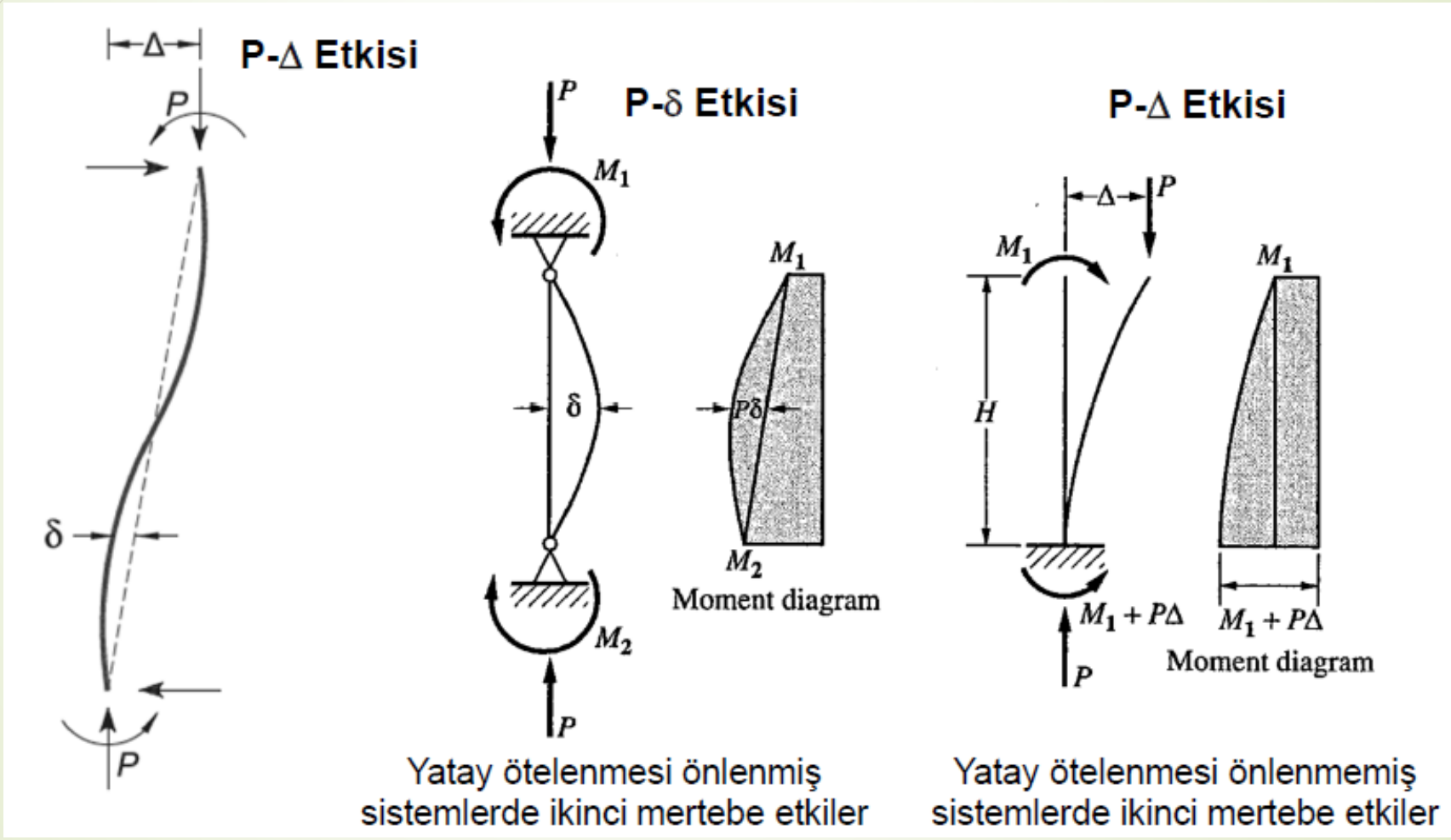
Q=15*1,2+40*1,6=82 kN

Mx=QL/4= 82*500/4=10250 kNcm olarak bulunur.

P/Pc=608/1200=0,51 > 0,2 olduğundan P/Pc+8/9(Mx/Mcx+My/Mcy) ≤ 1,0 geçerli olur.

0,51+8/9(0,50+0)=0,95 ≤ 1,0 yükler güvenli olarak taşınır.

İkinci Mertebe Etkiler



STABİLİTE TASARIMI/ DESIGN FOR STABILITY

Yapı sistemlerinin stabilite tasarımı, eleman bazındaki ve sistem genelindeki geometri değişimlerinin denge denklemlerine etkisi gözönüne alan ikinci mertebe teorisine göre analiz yapılmasını ve hesaplanan iç kuvvet büyüklüklerinin elemanların mevcut dayanımları ile karşılaştırılmasını öngörmektedir.

Yapı sistemlerinin stabilitesini etkileyen başlıca faktörler:

- Elemanların eğilme, kayma ve eksenel şekildeğiştirmeleri
- Yapı sisteminin yerdeğiştirmesinde etkili olan tüm şekildeğiştirmeler (kiriş-kolon birleşim bölgesi şekildeğiştirmeleri, vb.)
- Eleman şekildeğiştirmesine ait P-δ ikinci mertebe etkileri
- Sistem yerdeğiştirmesine ait P−Δ ikinci mertebe etkileri
- Geometrik ön kusurlar
- Doğrusal olmayan şekildeğiştirmeler ile dayanım ve rijitliklerdeki belirsizlikler

STABİLİTE TASARIMI/ DESIGN FOR STABILITY

İkinci mertebe teorisi doğrusal olmadığından süperpozisyon ilkesi geçerli değildir. Bu nedenle işletme (servis) yüklerinin ilgili yük birleşimleri için kendilerine ait yük katsayıları ile çarpımından oluşan toplam yükler altında, sistem ikinci mertebe teorisine göre hesaplanır. Toplam yükler YDKT yük birleşimlerinden elde edilen yüklemeden elde edilecektir.

Yapısal stabiliteyi dikkate alan üç adet ikinci mertebe analizi vardır.

- Genel (Doğrudan) Analiz Yöntemi
- Burkulma Boyu (Etkin Uzunluk) Yöntemi
- Yaklaşık İkinci Mertebe Analizi

STABİLİTE TASARIMI/ DESIGN FOR STABILITY

Yapı stabilitesini etkileyen faktörler-Geometrik Ön Kusurlar:

Geometrik ön kusurların yapısal stabiliteye etkileri doğrudan doğruya modellenebildiği gibi, fiktif kuvvetler kullanılarakta hesaba katılabilir.

Yapısal stabilite analiznde gözönüne alınan sistem ön kusurları, düğüm noktalarının konumundaki geometrik ön kusurlar olarak tanımlanmaktadır. Doğrusal elemanlarda uç noktalarını birleştiren doğrudan olan yerel sapmalar olarak tanımlanan eleman ön kusurları ise bu bölümün konusu dışında kalmaktadır.

GENERAL STABILITY REQUIREMENTS

Stability shall be provided for the structure as a whole and for each of its elements

The effects of all of the following on the stability of the structure and its elements shall be considered:

(a) flexural, shear and axial member deformations, and all other component and connection deformations that contribute to the displacements of the structure;

(b) second-order effects (including *P*-Δ and *P*-δ effects);

(c) Geometric imperfections;

(d) stiffness reductions due to inelasticity, including the effect of partial yielding of the cross section which may be accentuated by the presence of residual stresses; and

(e) uncertainty in system, member, and connection strength and stiffness.

All load-dependent effects shall be calculated at a level of loading corresponding to LRFD load combinations or 1.6 times ASD load combinations.

Analysis Methods of Design Regarding Second Order Effect

Direct Analysis Method of Design :

is permitted for all structures, and can be based on either elastic or inelastic analysis.

For design by elastic analysis, required strengths and available strengths are calculated.

Alternative Methods of Design :

The effective length method and the first-order analysis method,

based on elastic analysis and are permitted as alternatives to the direct analysis method for structures that satisfy the limitations specified

CALCULATION OF REQUIRED STRENGTHS

For the direct analysis method of design, the required strengths of components of the structure shall be determined from an elastic analysis. The analysis shall include consideration of initial imperfections.

General Analysis Requirements

The analysis of the structure shall conform to the following requirements:

(a) The analysis shall consider flexural, shear and axial member deformations, and all other component and connection deformations that contribute to displacements of the structure. The analysis shall incorporate reductions in all stiffnesses that are considered to contribute to the stability of the structure.

(b) The analysis shall be a second-order analysis that considers both P-Δ and P-δ effects, except that it is permissible to neglect the effect of P-δ on the response of the structure when the following conditions are satisfied:

(1) the structure supports gravity loads primarily through nominally vertical columns, walls or frames;

(2) the ratio of maximum second-order drift to maximum first-order drift (both determined for LRFD load combinations or 1.6 times ASD load combinations, with stiffnesses adjusted as specified in Section C2.3) in all stories is equal to or less than 1.7; and

(3) no more than one-third of the total gravity load on the structure is supported by columns that are part of moment-resisting frames in the direction of translation being considered. It is necessary in all cases to consider P-δ effects in the evaluation of individual members subject to compression and flexure.

c) The analysis shall consider all gravity and other applied loads that may influence the stability of the structure.

d) For design by LRFD, the second-order analysis shall be carried out under LRFD load combinations. For design by ASD, the second-order analysis shall be carried out under 1.6 times the ASD load combinations, and the results shall be divided by 1.6 to obtain the required strengths of components.

CALCULATION OF REQUIRED STRENGTHS

Consideration of Initial System Imperfections
The effect of initial imperfections in the position of points of intersection of members on the stability of the structure shall be taken into account either by direct modeling of these imperfections in the analysis

1. Direct Modeling of Imperfections
In all cases, it is permissible to account for the effect of initial system imperfections by including the imperfections directly in the analysis. The structure shall be analyzed with points of intersection of members displaced from their nominal locations. The magnitude of the initial displacements shall be the maximum amount considered in the design; the pattern of initial displacements shall be such that it provides the greatest destabilizing effect.

CALCULATION OF REQUIRED STRENGTHS

2. Use of Notional Loads to Represent Imperfections
For structures that support gravity loads primarily through nominally vertical columns, walls or frames, it is permissible to use notional loads to represent the effects of initial system imperfections in the position of points of intersection of members in accordance with the requirements of this section. The notional load shall be applied to a model of the structure based on its nominal geometry.

a) Notional loads shall be applied as lateral loads at all levels. The notional loads shall be additive to other lateral loads and shall be applied in all load combinations. The magnitude of the notional loads shall be:

$$N_i = 0.002\alpha Y_i \quad \text{(C2-1)}$$

where $\alpha = 1.0$ (LRFD); $\alpha = 1.6$ (ASD); N_i = notional load applied at level i, kips (N)
Y_i = gravity load applied at level i from the LRFD load combination or ASD load combination, as applicable, kips (N)

CALCULATION OF REQUIRED STRENGTHS

b) The notional load at any level, *Ni*, shall be distributed over that level in the same manner as the gravity load at the level. The notional loads shall be applied in the direction that provides the greatest destabilizing effect.

User Note: For most building structures, the requirement regarding notional load direction may be satisfied as follows: for load combinations that do not include lateral loading, consider two alternative orthogonal directions of notional load application, in a positive and a negative sense in each of the two directions, in the same direction at all levels; for load combinations that include lateral loading, apply all notional loads in the direction of the resultant of all lateral loads in the combination.

c) The notional load coefficient of 0.002 in Equation C2-1 is based on a nominal initial story out-of-plumbness ratio of 1/500; where the use of a different maximum out-of-plumbness is justified, it is permissible to adjust the notional load coefficient proportionally

CALCULATION OF REQUIRED STRENGTHS

d) For structures in which the ratio of maximum second-order drift to maximum first-order drift (both determined for LRFD load combinations or 1.6 times ASD load combinations) in all stories is equal to or less than 1.7, it is permissible to apply the notional load, *Ni*, only in gravity-only load combinations and not in combinations that include other lateral loads.

3. Adjustments to Stiffness

The analysis of the structure to determine the required strengths of components shall use reduced stiffnesses, as follows:

a) A factor of 0.80 shall be applied to all stiffnesses that are considered to contribute to the stability of the structure. It is permissible to apply this reduction factor to all stiffnesses in the structure.

b) An additional factor, τb, shall be applied to the flexural stiffnesses of all members whose flexural stiffnesses are considered to contribute to the stability of the structure. For noncomposite members, τb shall be defined as follows.

CALCULATION OF REQUIRED STRENGTHS

(1) When $\alpha P_r / P_{ns} \leq 0.5$

$\tau_b = 1.0$ (C2-2a)

(2) When $\alpha P_r / P_{ns} > 0.5$

$\tau_b = 4(\alpha P_r/P_{ns})[1- (\alpha P_r/P_{ns})]$ (C2-2b)

where $\alpha = 1.0$ (LRFD); $\alpha = 1.6$ (ASD); P_r = required axial compressive strength using LRFD or ASD load combinations, kips (N); P_{ns} = cross-section compressive strength; for nonslender-element sections, $P_{ns} = F_y A_g$, and for slender-element sections, $P_{ns} = F_y A_e$, where A_e is as defined in Section E7, kips (N)

CALCULATION OF REQUIRED STRENGTHS

User Note: Taken together, Sections (a) and (b) require the use of 0.8τ*b* times the nominal elastic flexural stiffness and 0.8 times other nominal elastic stiffnesses for structural steel members in the analysis.

c) In structures to which Section C2.2b is applicable, in lieu of using τ*b* < 1.0 where α*Pr*/*Pns* > 0.5, it is permissible to use τ*b* = 1.0 for all noncomposite members if a notional load of 0.001α*Yi* [where *Yi* is as defined in Section C2.2b(a)] is applied at all levels, in the direction specified in Section C2.2b(b), in all load combinations. These notional loads shall be added to those, if any, used to account for the effects of initial imperfections in the position of points of intersection of members and shall not be subject to the provisions of Section C2.2b(d).

d) (d) Where components comprised of materials other than structural steel are considered to contribute to the stability of the structure and the governing codes and specifications for the other materials require greater reductions in stiffness, such greater stiffness reductions shall be applied to those components

CALCULATION OF AVAILABLE STRENGTHS

For the direct analysis method of design, the available strengths of members and connections shall be calculated in accordance with the provisions of Chapters D through K, as applicable, with no further consideration of overall structure stability. The effective length for flexural buckling of all members shall be taken as the unbraced length unless a smaller value is justified by rational analysis.

Bracing intended to define the unbraced lengths of members shall have sufficient stiffness and strength to control member movement at the braced points.

İkinci mertebe moment hesabı

$$M_r = B_1 M_{nt} + B_2 M_{\ell t}$$

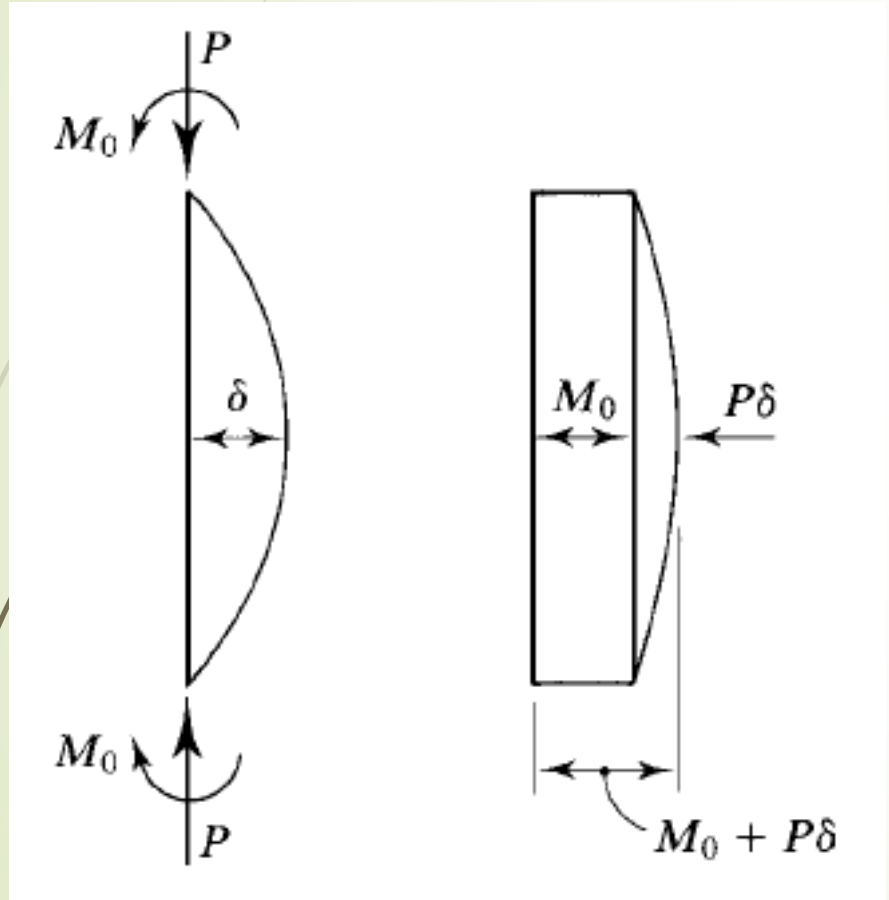
- *Mr* = gerekli moment değeri
  = *Mu* for LRFD
- *Mnt* =Yanal olarak desteklenmiş sistemde oluşan moment değeri (nt indisi "no translation").
- *Mlt* = Yanal olarak desteklenmemiş sistemde oluşan moment değeri (lt indisi "lateral translation"). Bu moment değeri yatay yük veya dengelenmemiş yerçekimi yüküne neden olabilir. Yanal olarak desteklenmiş elemanda *Mlt değeri sıfırdır. Yatay yükden dolayı oluşan moment değeri varsa Mlt değeri hesaplanması gerekebilir.*
- *B*1 = yanal olarak desteklenmiş elemanda hesaplanan artım katsayısı (*P-d* moments).
- *B*2 = yanal olarak desteklenmemiş elemanda hesaplanan artım katsayısı (*P*-Δ moments).

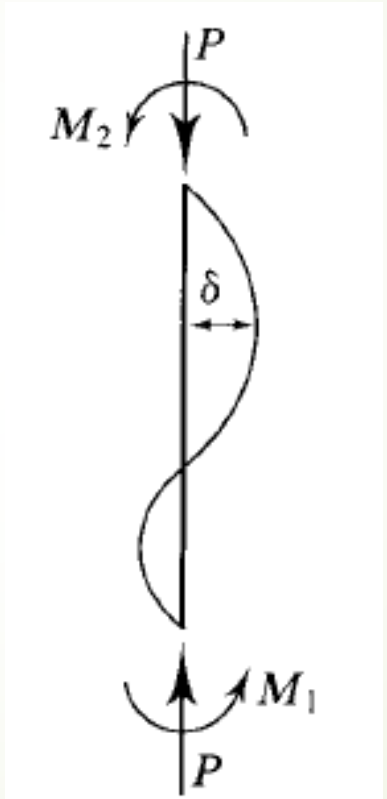
İkinci mertebe eksenel kuvvet hesabı

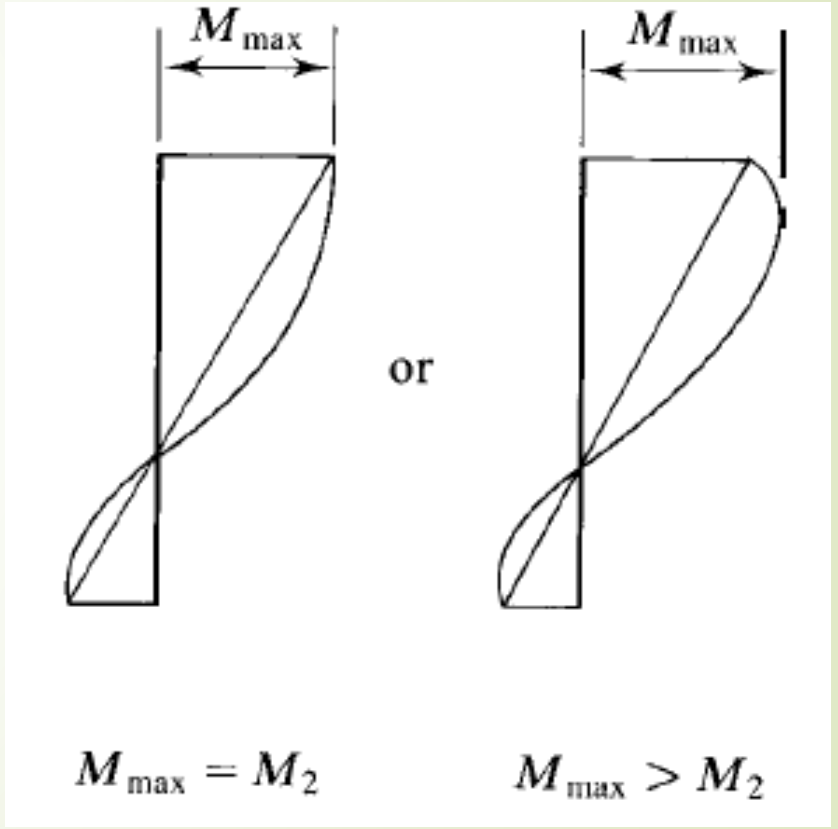
$$P_r = P_{nt} + B_2 P_{\ell t}$$

- P_{nt} = Yanal olarak desteklenmiş sistemde oluşan eksenel yük değeri
- P_{lt} = Yanal olarak desteklenmemiş sistemde oluşan eksenel yük değeri

Ötelenmesi Engellenmiş







$$B_1 = \frac{C_m}{1-(\alpha P_r / P_{e1})} \geq 1$$

P_r = required unamplified axial compressive strength ($P_{nt} + P_{\ell t}$)
= P_u for LRFD
= P_a for ASD

α = 1.00 for LRFD
= 1.60 for ASD

$$P_{e1} = \frac{\pi^2 EI^*}{(K_1 L)^2} \qquad \text{(AISC Equation A-8-5)}$$

EI^* = flexural rigidity

In the direct analysis method, EI^* is a reduced stiffness obtained as

$$EI^* = 0.8\tau_b EI \qquad (6.8)$$

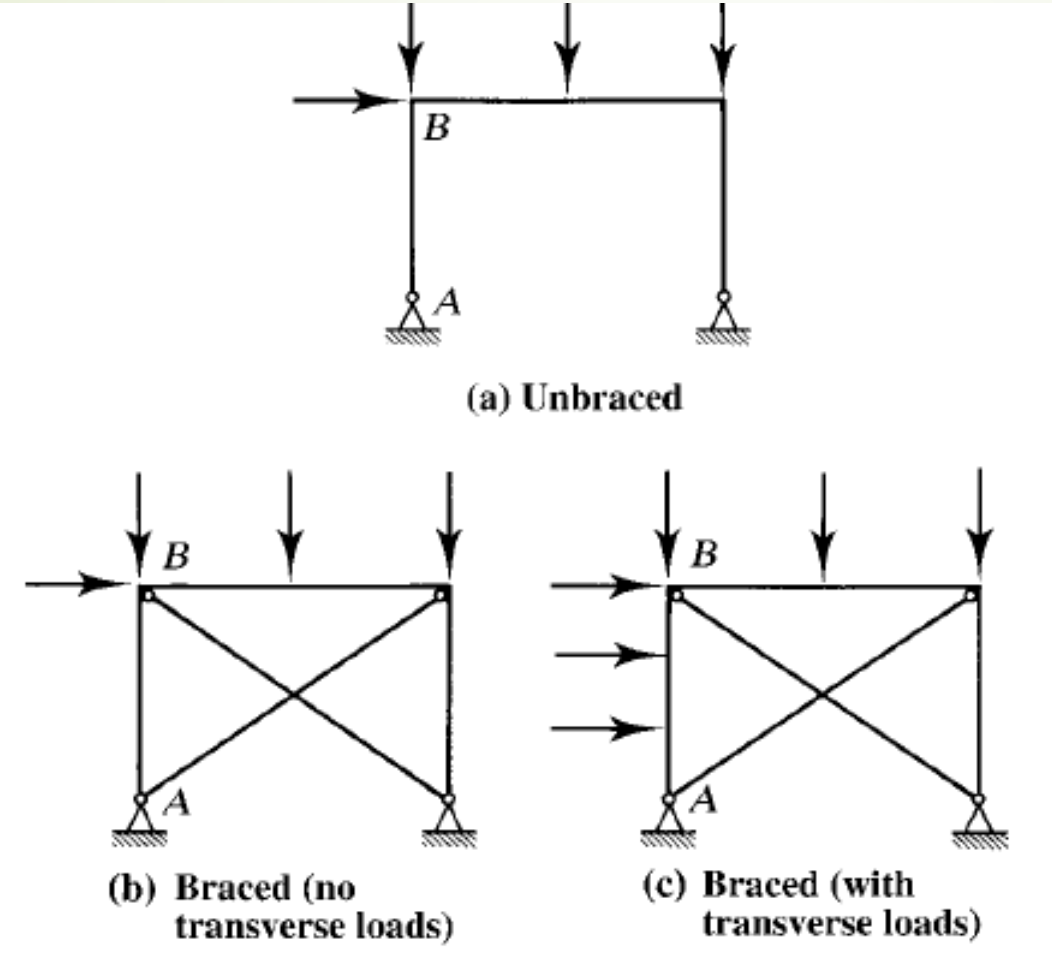
where

τ_b = a stiffness reduction factor

$$= 1.0 \text{ when } \frac{P_r}{P_y} \leq 0.5 \qquad \text{(AISC Equation C2-2a)}$$

$$= 4\left(\frac{P_r}{P_y}\right)\left(1-\frac{P_r}{P_y}\right) \text{ when } \frac{P_r}{P_y} > 0.5 \qquad \text{(AISC Equation C2-2b)}$$

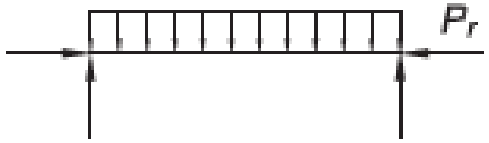
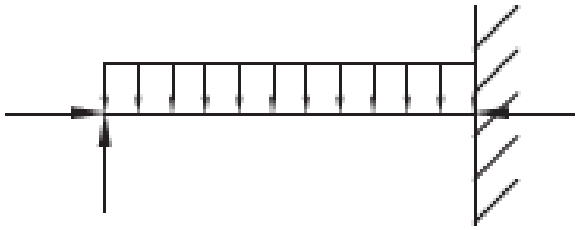
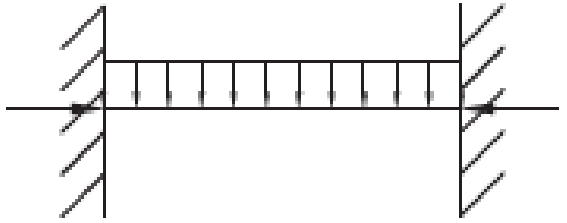
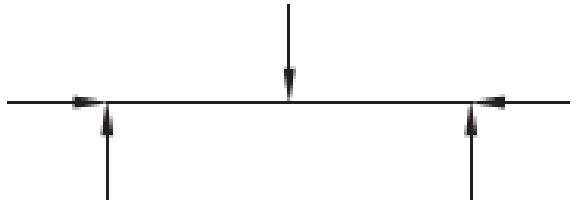
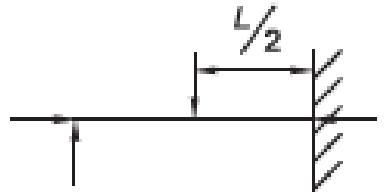
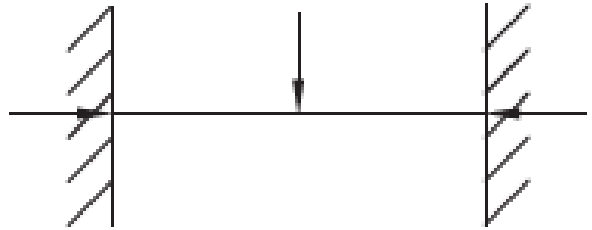
Cm katsayısı



$$C_m = 0.6 - 0.4\left(\frac{M_1}{M_2}\right)$$

$$C_m = 1 + \Psi\left(\frac{\alpha P_r}{P_{e1}}\right)$$

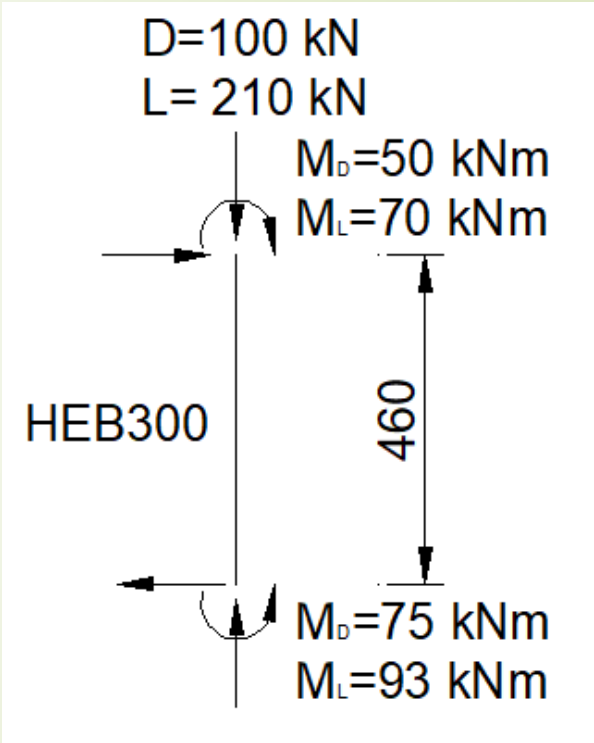
TABLE C-A-8.1
Amplification Factors ψ and C_m

Case	ψ	C_m
	0	1.0
	−0.4	$1-0.4\frac{\alpha P_r}{P_{e1}}$
	−0.4	$1-0.4\frac{\alpha P_r}{P_{e1}}$
	−0.2	$1-0.2\frac{\alpha P_r}{P_{e1}}$
	−0.3	$1-0.3\frac{\alpha P_r}{P_{e1}}$
	−0.2	$1-0.2\frac{\alpha P_r}{P_{e1}}$

Örnek

Şekilde görülen elemana gelen yükler verilmiştir. Burkulma katsayıları yöntemine göre analiz edilecek elemanda çelik malzemesi olarak S355 kullanıldığı düşünülürse şartnameye göre profilin yükler altında güvenli olduğunu kontrol ediniz.

(burkulma boyu katsayısı Kx=Ky=1)



Yük katsayıları ile çarpılmış yükler (1.2D+1.6L)

1.2x100+1.6x210=456 kN

Yük katsayıları ile çarpılmış moment değeleri (1.2D+1.6L)

1.2x50+1.6x70=172 kNm

1.2x75+1.6x93=238,8 kNm

Pr=Pu=Pnt+β_2xPlt=456+0=456 kN (β_2 =0 çaprazlı sistemlerde)

Basınç çubukları konusundan HEB300 profilinin (Kx=Ky=1) 460 cm serbest uzunluğundaki durumu için taşıma kapasitesi

- *Güçlü eksen doğrultusunda 0,9*4816=4334 kN*
- *Zayıf eksen doğrultusunda 0,9*4012= 3610 kN olarak bulunur.*

Bu durumda profilin taşıma kapasitesi 3610 kN olarak bulunmuş olur.

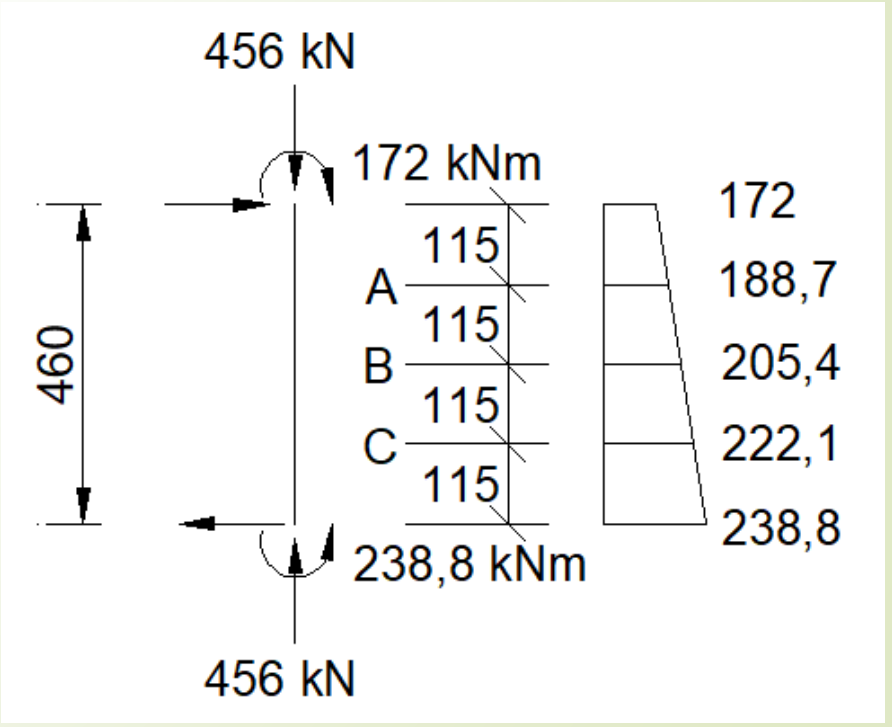
Kolon temiz açıklığı (desteklenmemiş uzunluk) 460 cm olduğundan

$$C_b = \frac{12{,}5 * 238{,}8}{2{,}5 * 238{,}8 + 3 * 188{,}7 + 4 * 205{,}4 + 3 * 222{,}1} = 1{,}13$$

$$C_m = 0{,}6 - 0{,}4 * \frac{M1}{M2} = 0{,}6 - 0{,}4 * \frac{172}{238{,}8} = 0{,}888$$

$$Pel = \frac{\pi^2 E I_x}{(K_x * L)^2} = \frac{\pi^2 * 20000 * 25170}{(1 * 460)^2} = 23456\ kN$$

$$\beta_1 = \frac{Cm}{1 - aPr/Pel} = \frac{0{,}888}{1 - 456/23456} = 0{,}905 < 1$$



Eğilme Momenti hesabından HEB300 için Moment kapasitesi 460 cm uzunluk için Cb=1,13

- Akma durumu dikkate alındığında 0,9*663= 596,7 kNm
- Yanal burulmalı burkulma durumu için 0,9*707= 636,3 kNm bulunur.

Bu durumda profilin güçlü eksen doğrultusundaki eğilme kapasitesi

Mcx=596,7 kNm olarak bulunmuş olur.

Mnt=238,8 kNm Mlt=0

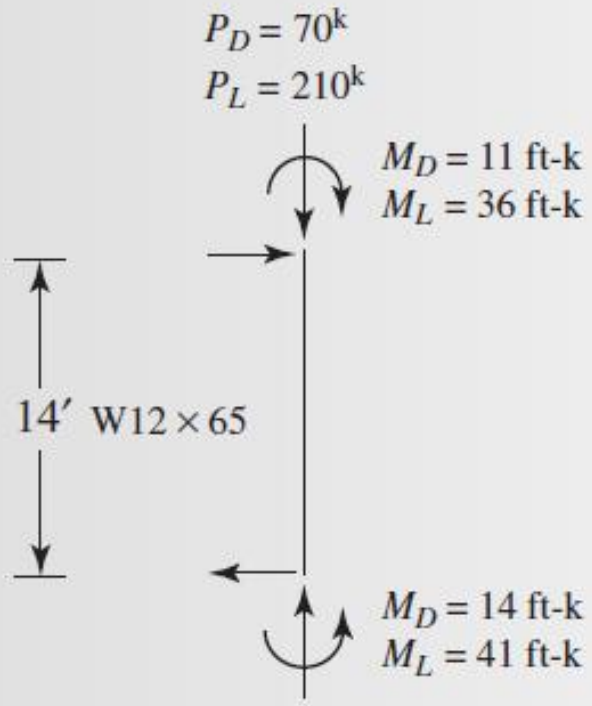
Mr= 1*Mnt+0*Mlt=238,8 kNm

P/Pc=456/3610=0,13 < 0,2 olduğundan P/2*Pc+(Mx/Mcx+My/Mcy) ≤ 1,0 geçerli olur.

0,13/2+(0,40+0)=0,46 ≤ 1,0 yükler güvenli olarak taşınır.

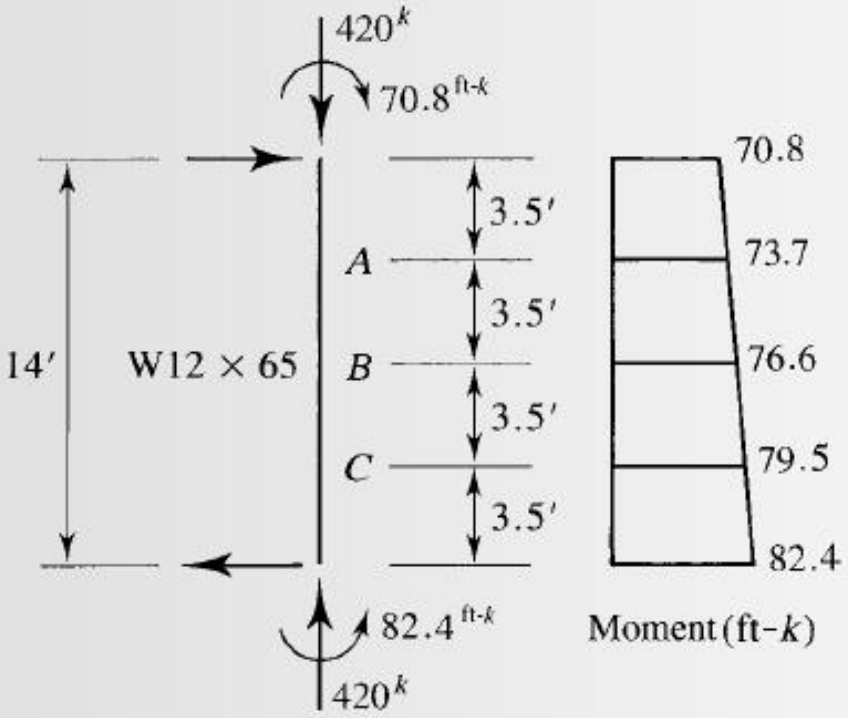
Örnek

The member shown in Figure 6.10 is part of a braced frame. An analysis consistent with the effective length method was performed; therefore, the flexural rigidity, EI, was unreduced. If A572 Grade 50 steel is used, is this member adequate? $K_x = K_y = 1.0$.



The factored loads, computed from load combination 2, are shown in Figure 6.11. Determine which interaction formula to apply. The required compressive strength is

$$P_r = P_u = P_{nt} + B_2 P_{\ell t} = 420 + 0 = 420 \text{ kips} \qquad (B_2 = 0 \text{ for a braced frame})$$



From the column load tables, for $KL = 1.0 \times 14 = 14$ feet, the axial compressive strength of a W12 × 65 is

$$\phi_c P_n = 685 \text{ kips}$$

$$\frac{P_u}{\phi_c P_n} = \frac{420}{685} = 0.6131 > 0.2 \qquad \therefore \text{ Use Equation 6.3 (AISC Equation H1-1a).}$$

In the plane of bending,

$$P_{e1} = \frac{\pi^2 EI}{(K_1 L)^2} = \frac{\pi^2 EI_x}{(K_x L)^2} = \frac{\pi^2 (29{,}000)(533)}{(1.0 \times 14 \times 12)^2} = 5405 \text{ kips}$$

$$C_m = 0.6 - 0.4\left(\frac{M_1}{M_2}\right) = 0.6 - 0.4\left(-\frac{70.8}{82.4}\right) = 0.9437$$

$$B_1 = \frac{C_m}{1 - (\alpha P_r / P_{e1})} = \frac{C_m}{1 - (1.00 P_u / P_{e1})} = \frac{0.9437}{1 - (420/5405)} = 1.023$$

From the Beam Design Charts in Part 3 of the *Manual* with $C_b = 1.0$ and $L_b = 14$ feet, the moment strength is

$$\phi_b M_n = 345 \text{ ft-kips}$$

For the actual value of C_b, refer to the moment diagram of Figure 6.11:

$$C_b = \frac{12.5 M_{\max}}{2.5 M_{\max} + 3M_A + 4M_B + 3M_C}$$

$$= \frac{12.5(82.4)}{2.5(82.4) + 3(73.7) + 4(76.6) + 3(79.5)} = 1.060$$

$$\therefore \quad \phi_b M_n = C_b(345) = 1.060(345) = 366 \text{ ft-kips}$$

But $\phi_b M_p = 356$ ft-kips (from the charts) < 366 ft-kips $\quad \therefore$ Use $\phi_b M_n = 356$ ft-kips.

(Since a W12×65 is noncompact for $F_y = 50$ ksi, 356 ft-kips is the design strength based on FLB rather than full yielding of the cross section.) The factored load moments are

$$M_{nt} = 82.4 \text{ ft-kips} \quad M_{\ell t} = 0$$

From AISC Equation A-8-1, the required moment strength is

$$M_r = M_u = B_1 M_{nt} + B_2 M_{\ell t} = 1.023(82.4) + 0 = 84.30 \text{ ft-kips} = M_{ux}$$

From Equation 6.3 (AISC Equation H1-1a),

$$\frac{P_u}{\phi_c P_n} + \frac{8}{9}\left(\frac{M_{ux}}{\phi_b M_{nx}} + \frac{M_{uy}}{\phi_b M_{ny}}\right) = 0.6131 + \frac{8}{9}\left(\frac{84.30}{356} + 0\right) = 0.824 < 1.0 \qquad \text{(OK)}$$

The member is satisfactory.

Ötelenmesi Engellenmemiş

$$B_2 = \frac{1}{1 - \frac{\alpha P_{story}}{P_{e\,story}}} \geq 1 \qquad \text{(AISC Equation A-8-6)}$$

where

$\alpha = 1.00$ for LRFD
$= 1.60$ for ASD

P_{story} = sum of required load capacities for all columns in the story under consideration (factored for LRFD, unfactored for ASD)

$P_{e\,story}$ = total elastic buckling strength of the story under consideration

This story buckling strength may be obtained by a sidesway buckling analysis or as

$$P_{e\,story} = R_M \frac{HL}{\Delta_H} \qquad \text{(AISC Equation A-8-7)}$$

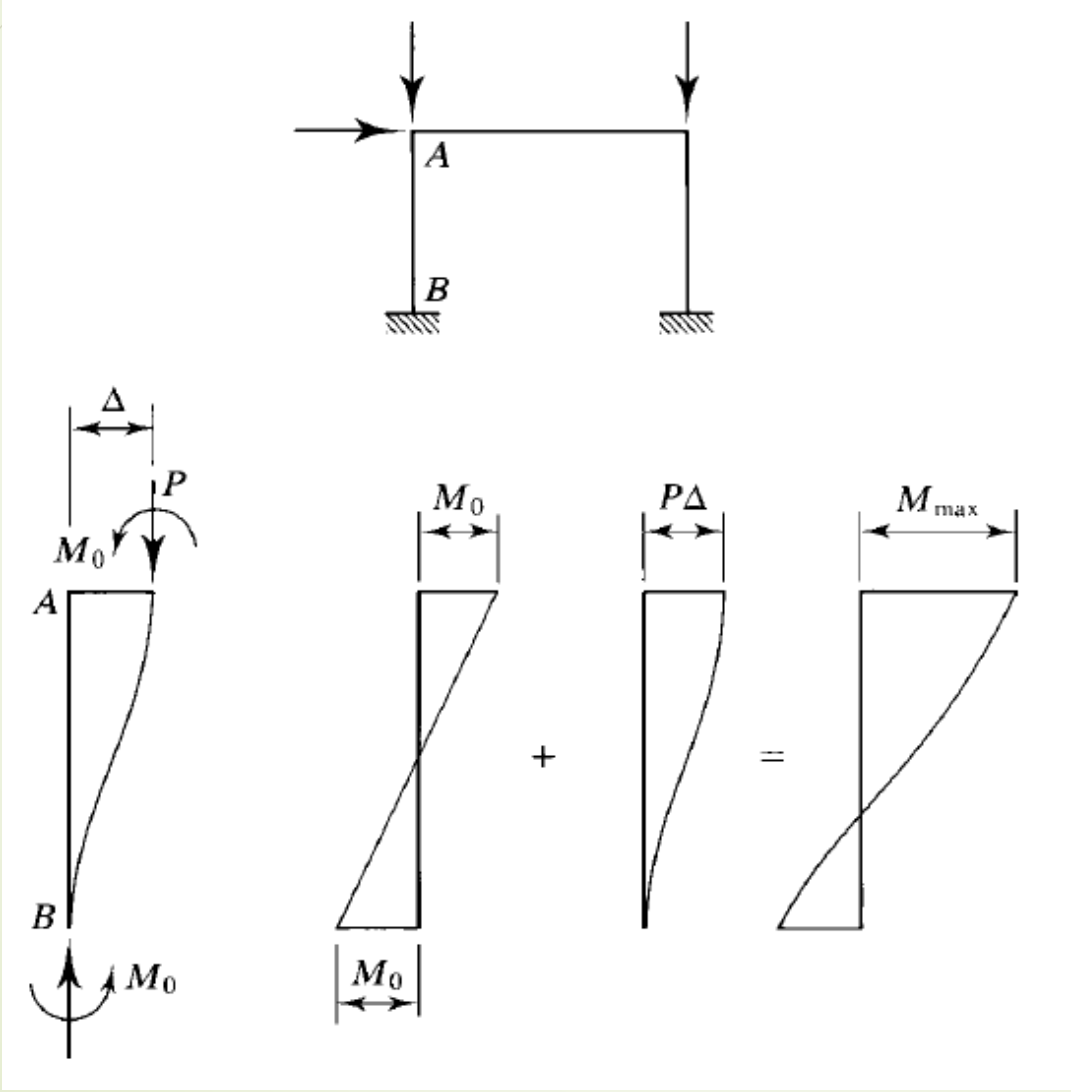
$$R_M = 1 - 0.15\frac{P_{mf}}{P_{\text{story}}}$$ (AISC Equation A-8-8)

P_{mf} = sum of vertical loads in all columns in the story that are part of *moment frames*

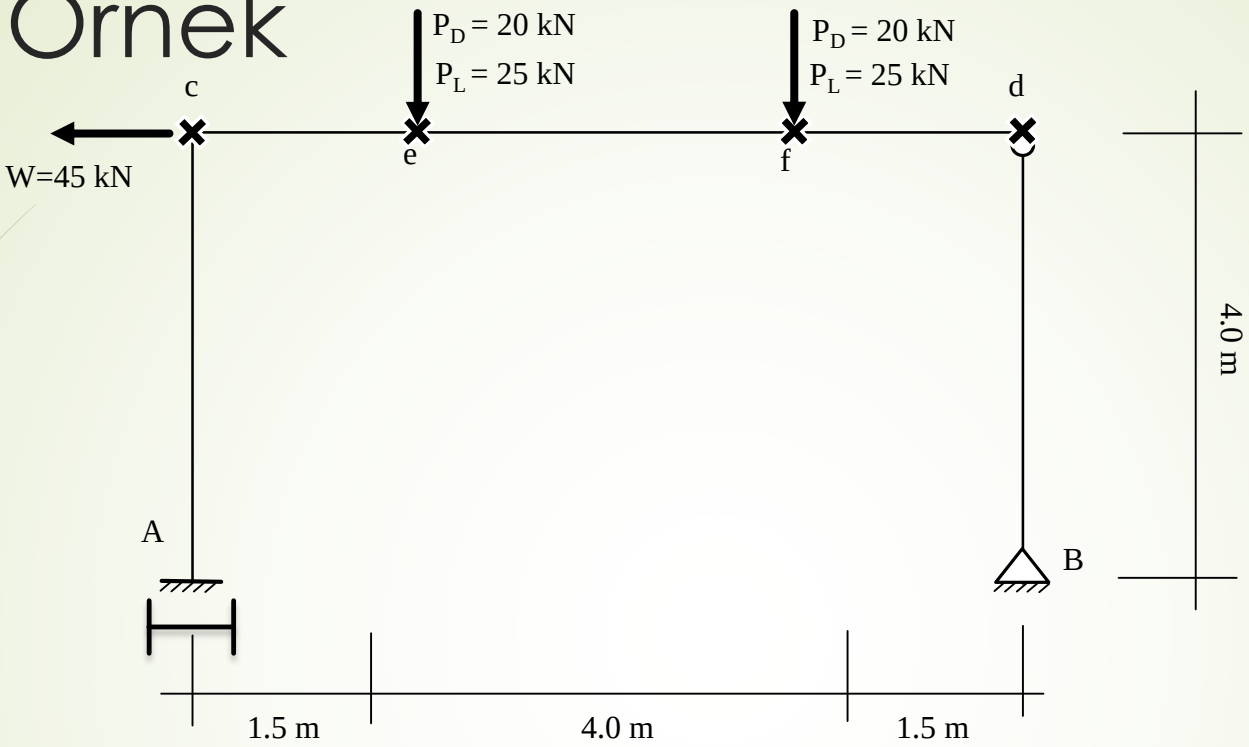
L = story height

Δ_H = interstory drift = drift (sidesway displacement) of the story under consideration

H = story shear = sum of all horizontal forces causing $_H$


A
B
Δ
P
M0
A
B
M0
M0
PΔ
Mmax
+
=
M0

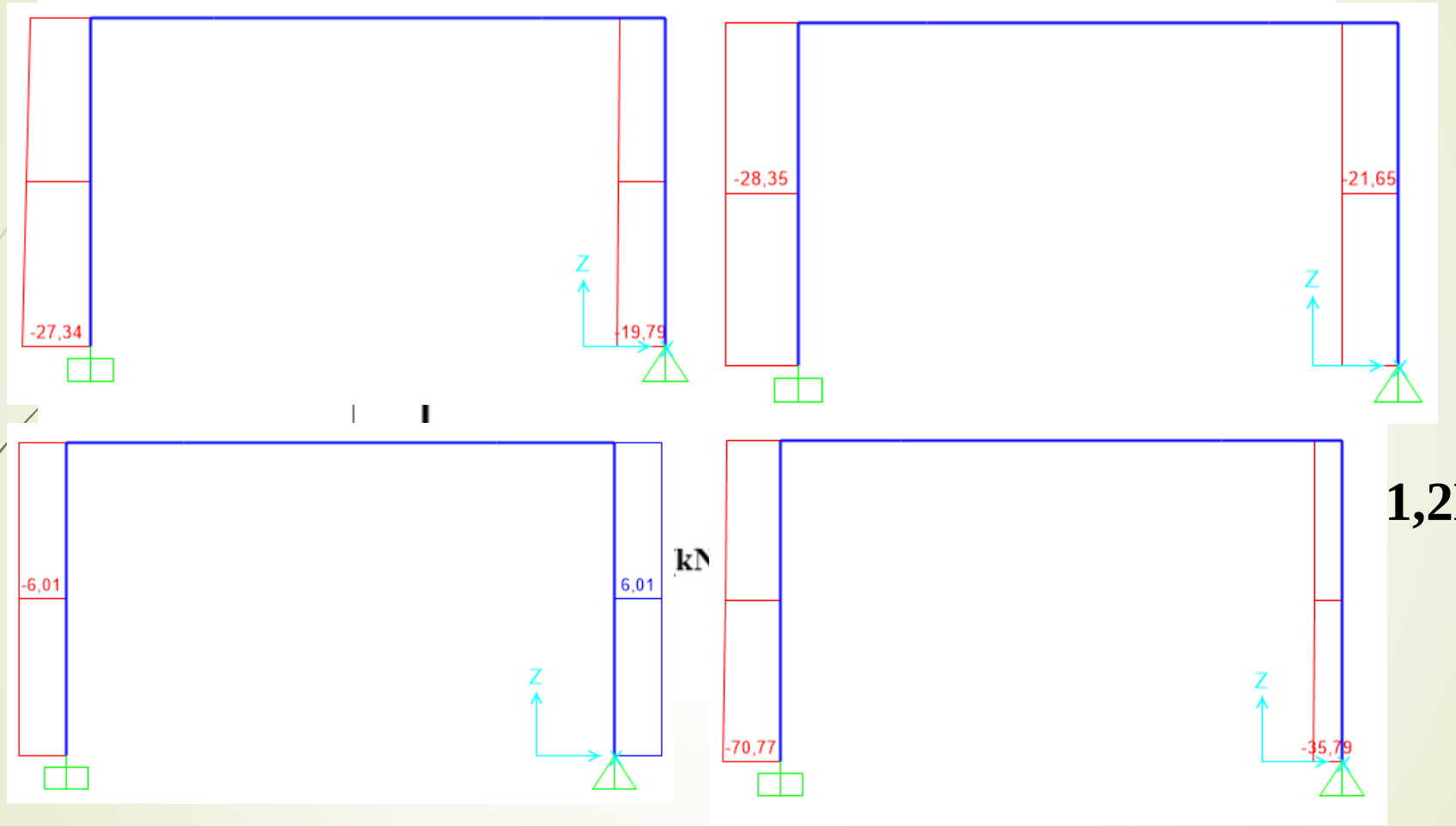
Örnek



Şekilde görülen sistemde **A-c kolonu HE240B**, **c-d kirişi IPE270** ve **d-B kolonu HE180A** profillerden imal edilmiştir. Tüm elemanlarda **S355** yapı çeliği kullanılmıştır. A ve c noktalarından dik düzlemde mafsallar ile tutulan A-c kolonunu, verilen dış yükler için 1,2D+L+1,6W yük kombinasyonunu kullanarak kontrol ediniz. (1,6W için Δ_H=43,68 mm)

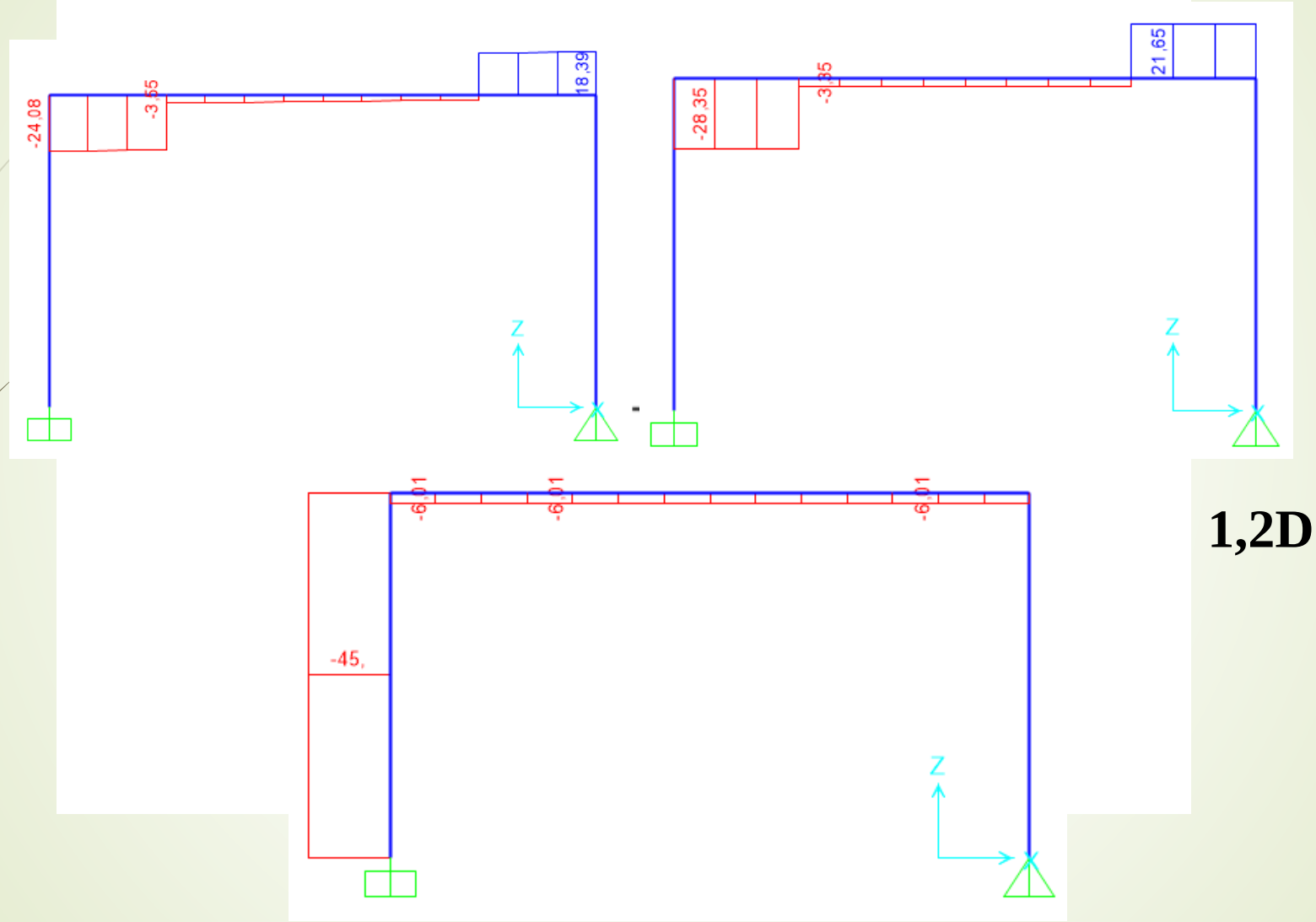
KESİT	ÖLÇÜLER						STATİK DEĞERLER										
							KUVVETLİ EKSEN x-x					ZAYIF EKSEN y-y					
	d	b	t_w	t_f	A	h	I_x	$W_{el.x}$	$W_{pl.x}$	ix	A_{vx}	Iy	$W_{el.y}$	$W_{pl.y}$	iy	J	C_w
	mm	mm	mm	mm	cm^2	mm	cm^4	cm^3	cm^3	cm	cm^2	cm^4	cm^3	cm^3	cm	cm^4	cm^6
HE240B	240	240	10,0	17,0	106,00	164	11260,0	938,3	1053,0	10,31	33,23	3923	326,9	498,4	6,08	102,70	486900

Normal Kuvvet Diyagramları

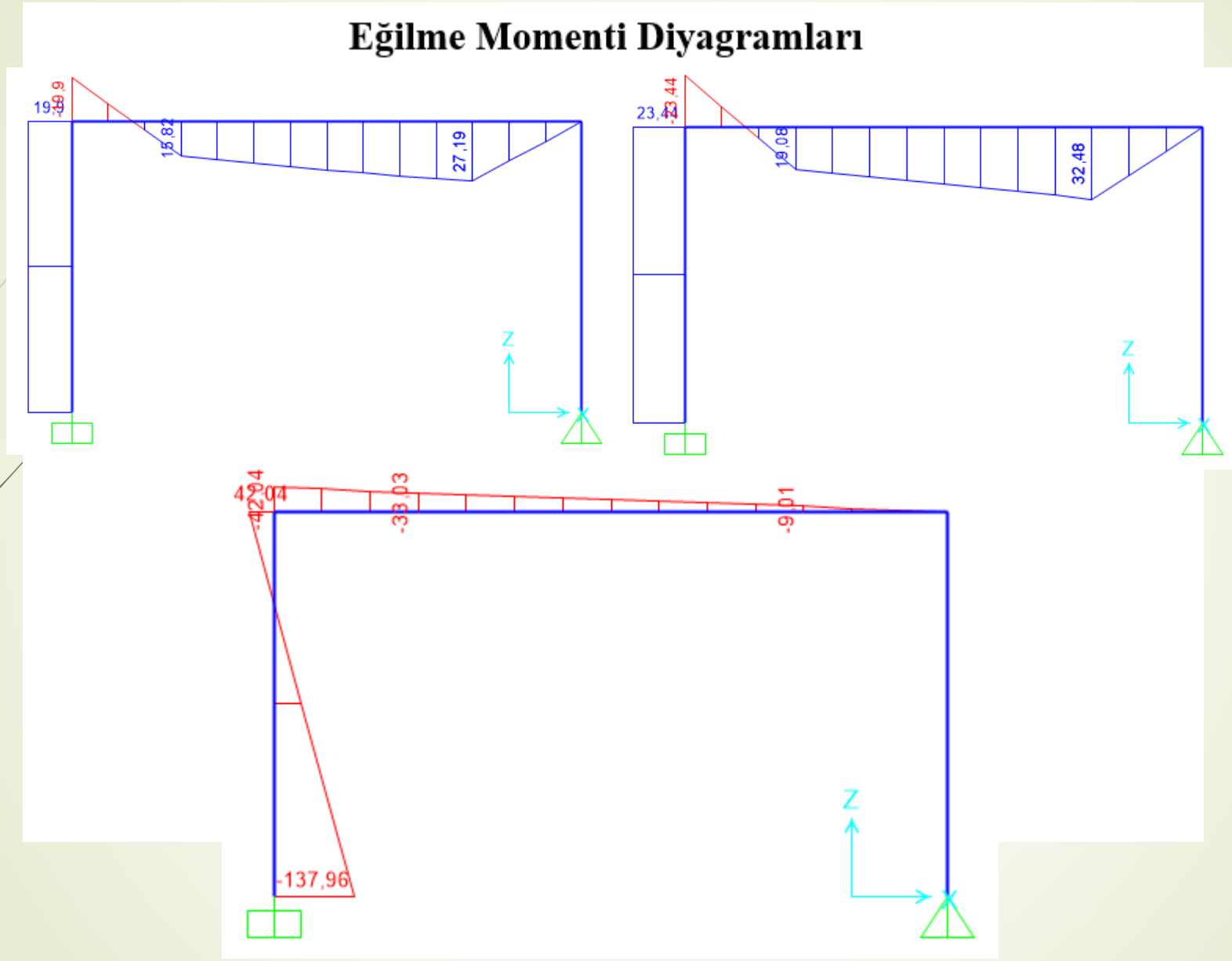


1,2D+L+1,6W

Kesme Kuvveti Diyagramları



1,2D+L+1,6W


Eğilme Momenti Diyagramları
-19,9
15,82
27,19
Z
-23,44
18,08
32,48
Z
-42,04
-33,03
-9,01
-137,96
Z

1,2D+L+1,6W

Basınç Dayanımı

Yerel burkulma kontrolü

$$\lambda = \frac{b}{2t_f} = \frac{240}{2 \cdot 17} = 7{,}06 < \lambda_r = 0{,}56\sqrt{\frac{E}{F_y}} = 13{,}29$$ Başlık narin değil

$$\lambda = \frac{h}{t_w} = \frac{164}{10} = 16{,}4 < \lambda_r = 1{,}49\sqrt{\frac{E}{F_y}} = 35{,}37$$ Gövde narin değil

Burkulma boyu katsayıları

$$G = \frac{\sum\left(E_c I_c / L_c\right)}{\sum\left(E_g I_g / L_g\right)} \qquad G_A = \frac{11260/400}{5790/700} = 3{,}40 \qquad G_B = 1$$

$$K_x = \sqrt{\frac{3{,}40 \cdot (1{,}6 \cdot 1 + 4) + (4 \cdot 1 + 7{,}5)}{3{,}40 + 1 + 7{,}5}} = 1{,}602 \qquad K_y = 1$$

$$\left.\begin{aligned}\lambda_x &= \frac{1{,}602 \cdot 400}{10{,}31} = 62{,}15 \\ \lambda_y &= \frac{1 \cdot 400}{6{,}08} = 65{,}79\end{aligned}\right\} \quad \lambda = 65{,}79 < 4{,}71\sqrt{\frac{E}{F_y}} = 111{,}79$$

$$F_e = \frac{\pi^2 20000}{65{,}79^2} = 45{,}56\ k\,N/cm^2 \qquad F_{cr} = 0{,}658^{\frac{35{,}5}{45{,}56}} \cdot 35{,}5 = 25{,}62\,k\,N/cm^2$$

$$P_n = 106 \cdot 25{,}62 = 2717\ kN \qquad P_c = \phi P_n = 2445\ kN$$

Eğilme Dayanımı

Yerel burkulma kontrolü

$$\lambda = \frac{b}{2t_f} = \frac{240}{2 \cdot 17} = 7{,}06 < \lambda_r = 0{,}38\sqrt{\frac{E}{F_y}} = 9{,}02$$ Başlık kompakt

$$\lambda = \frac{h}{t_w} = \frac{164}{10} = 16{,}4 < \lambda_r = 3{,}76\sqrt{\frac{E}{F_y}} = 89{,}25$$ Gövde kompakt

$$M_p = F_y W_{px} = 35{,}5 \cdot 1053 = 37381{,}5\ kNcm$$

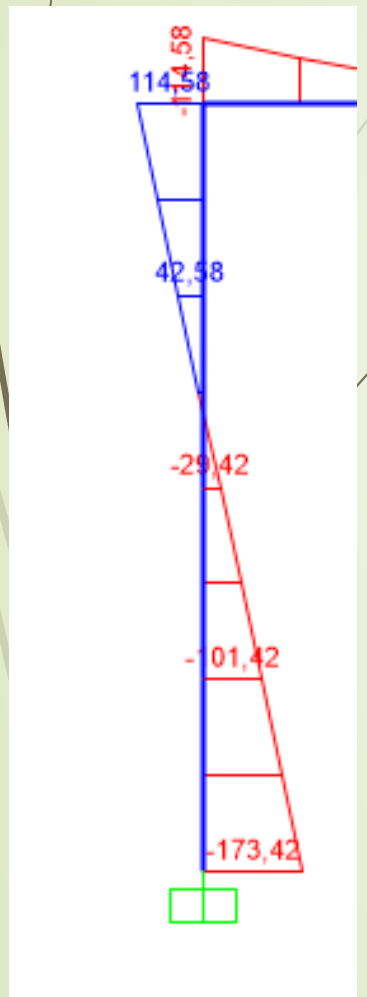
$$L_b = 400\ cm \qquad L_p = 1{,}76 i_y \sqrt{\frac{E}{F_y}} = 1{,}76 \cdot 6{,}08 \sqrt{\frac{20000}{35{,}5}} = 253{,}99\ cm$$

$$L_r = 1{,}95 i_{ts} \frac{E}{0{,}7F_y} \sqrt{\frac{Jc}{W_{ex}h_o} + \sqrt{\left(\frac{Jc}{W_{ex}h_o}\right)^2 + 6{,}76\left(\frac{0{,}7F_y}{E}\right)^2}}$$

$$i_{ts} = \sqrt{\frac{\sqrt{I_y C_w}}{W_{ex}}} = \sqrt{\frac{\sqrt{3923 \cdot 486900}}{938{,}3}} = 6{,}82\ cm \qquad c = 1$$

$$h_o = 24 - 1{,}7 = 22{,}3\ cm$$

$$L_r = 1{,}95 \cdot 6{,}82 \frac{20000}{0{,}7 \cdot 35{,}5} \sqrt{\frac{102{,}7 \cdot 1}{938{,}3 \cdot 22{,}3} + \sqrt{\left(\frac{102{,}7 \cdot 1}{938{,}3 \cdot 22{,}3}\right)^2 + 6{,}76\left(\frac{0{,}7 \cdot 35{,}5}{20000}\right)^2}} = 1111{,}52\ cm$$



$$1{,}2 \cdot (-19{,}9) + (-23{,}44) + 1{,}6 \cdot (-42{,}04) = -114{,}58\ kNm$$

$M_A = -42{,}58$

$M_B = 29{,}42$

$M_C = 101{,}42$

$$1{,}2 \cdot (-19{,}9) + (-23{,}44) + 1{,}6 \cdot (137{,}96) = 173{,}42\ kNm$$

$$C_b = \frac{12{,}5 \cdot 173{,}42}{2{,}5 \cdot 173{,}42 + 3 \cdot 42{,}58 + 4 \cdot 29{,}42 + 3 \cdot 101{,}42} = 2{,}20$$

$$M_n = C_b \left(M_p - \left(M_p - 0{,}7 F_y W_{ex}\right) \left(\frac{L_b - L_p}{L_r - L_p} \right) \right)$$

$$M_n = 2{,}20 \left(37381{,}5 - (37381{,}5 - 0{,}7 \cdot 35{,}5 \cdot 938{,}3) \left(\frac{400 - 253{,}99}{1111{,}52 - 253{,}99} \right) \right)$$

$$M_n = 76271{,}06\ kNcm > M_p$$

$$M_n = 37381{,}5\ kNcm$$

$$M_{cx} = 0{,}9 \cdot 37381{,}5 = 33643{,}35\ kNcm$$

$$P_{mf} = 27{,}34 + 28{,}35 + 6{,}01 = 61{,}70\ kN$$

$$P_{kat} = 1{,}2 \cdot (27{,}34 + 19{,}79) + (28{,}35 + 21{,}65) + 1{,}6 \cdot (6{,}01 - 6{,}01) = 106{,}56\ kN$$

$$R_M = 1 - 0{,}15 \cdot (61{,}70/106{,}56) = 0{,}913 \qquad H = 1{,}6 \cdot 45 = 72\ kN$$

$$\Delta_H = 5{,}57\ cm \qquad L = 400\ cm$$

$$P_{e,kat} = 0{,}913 \cdot \frac{72 \cdot 400}{4{,}368} = 6019{,}78\ kN$$

$$B_2 = \frac{1}{1 - \frac{1 \cdot 106{,}56}{6019{,}78}} = 1{,}018$$

$$P_{nt} = 1{,}2 \cdot 27{,}34 + 28{,}35 = 61{,}16\ kN$$
$$P_{lt} = 1{,}6 \cdot 6{,}01 = 9{,}62\ kN$$
$$P_r = 61{,}16 + 1{,}018 \cdot 9{,}62 = 70{,}95\ kN$$

$$C_m = 0{,}6 - 0{,}4\left(-\frac{114{,}58}{173{,}42}\right) = 0{,}86$$

$$P_{el} = \frac{\pi^2 20000 \cdot 11260}{(1 \cdot 400)^2} = 13891{,}47\ kN$$

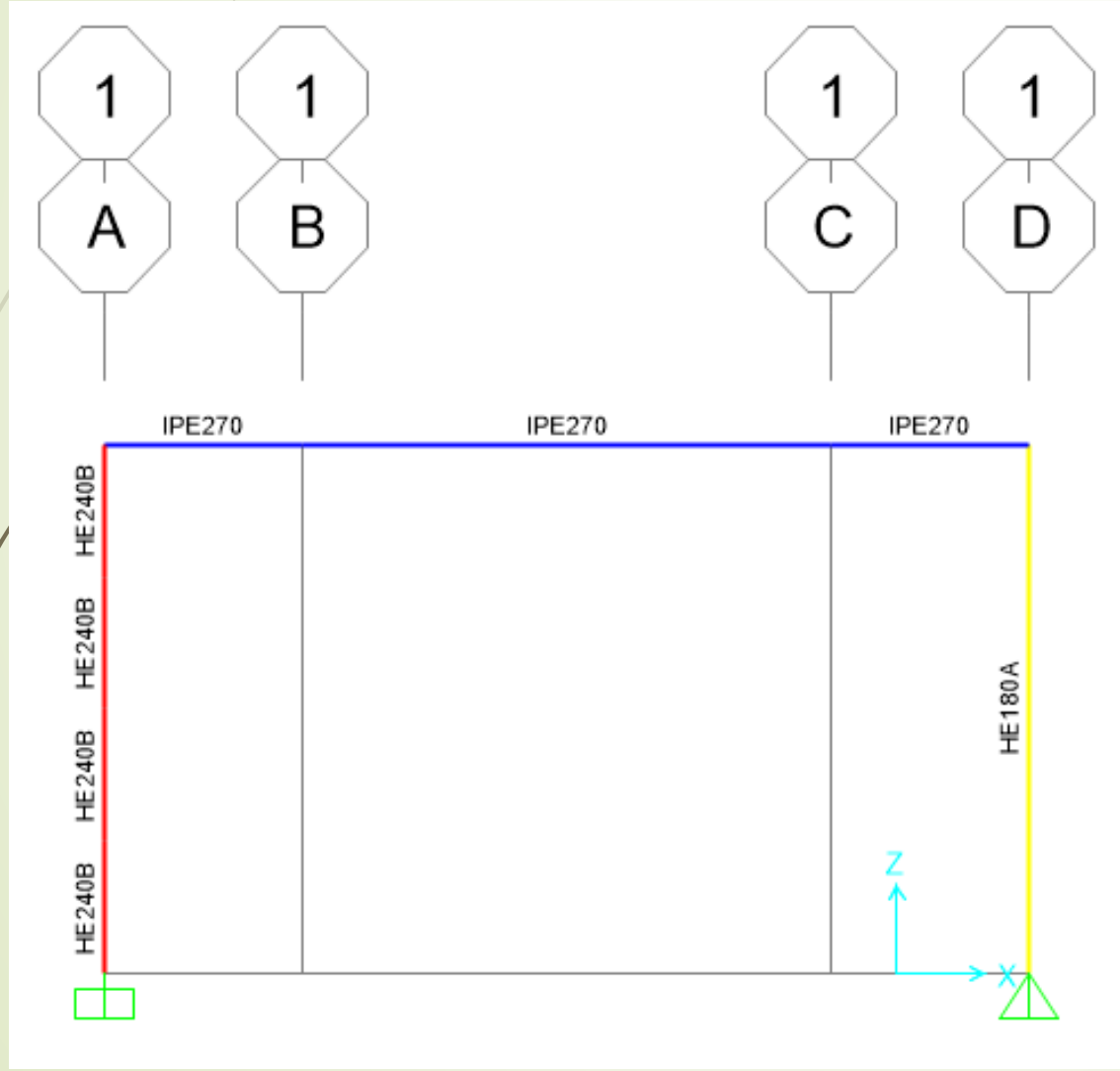
$$B_1 = \frac{0{,}86}{1 - \frac{1 \cdot 70{,}95}{13891{,}47}} = 0{,}864 < 1 \qquad \rightarrow \qquad B_1 = 1$$

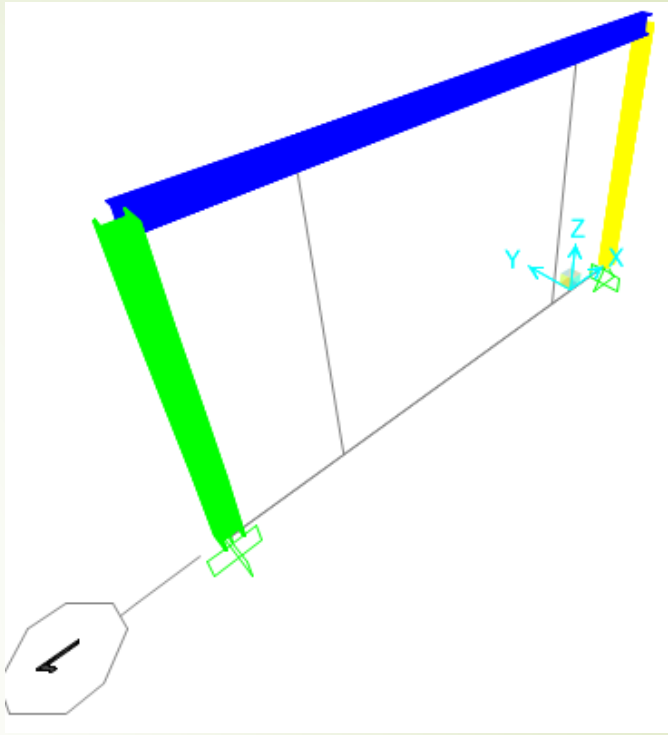
$$M_{nt} = 1{,}2 \cdot 19{,}9 + 23{,}44 = 47{,}32 \text{ kNm } (negatif)$$
$$M_{lt} = 1{,}6 \cdot 42{,}04 = 67{,}26\ kNm\ (negatif)\ (c - noktası)$$
$$M_{lt} = 1{,}6 \cdot 137{,}96 = 220{,}74\ kNm\ (pozitif)\ (A - noktası)$$

$$M_{rx} = 1 \cdot (-47{,}32) + 1{,}018 \cdot (-67{,}26) = 115{,}79\ kNm\ (mutlak\ değerce)$$
$$M_{rx} = 1 \cdot (-47{,}32) + 1{,}018 \cdot (220{,}74) = 177{,}39\ kNm$$

$$\frac{P_r}{P_c} = \frac{70{,}95}{2445} = 0{,}0290 < 0{,}2$$

$$\frac{0{,}0290}{2} + \frac{177{,}39}{336{,}43} = 0{,}54 < 1 \text{ (güvenlidir)}$$


1
1
1
1
A
B
C
D
IPE270
IPE270
IPE270
HE240B
HE240B
HE240B
HE240B
HE180A
Z
X


Z
Y
X


0,834
0,838
0,713
0,405
0,271
0,315
0,529
0,594
Z
X

P-M-M Ratio

AISC 360-16 STEEL SECTION CHECK (Summary for Combo and Station)
Units : KN, m, C

Frame :	11	X Mid:	-6,	Combo:	1,2d+l+l,6w
Length:	1,	Y Mid:	0,	Shape:	HE240B
Loc :	0,	Z Mid:	0,5	Class:	Compact

Design Type: Column
Frame Type: SCBF
Princpl Rot: 0, degrees

Provision: LRFD Analysis: Effective Length
D/C Limit=1, 2nd Order: General 2nd Order

PhiB=0,9	PhiC=0,9	PhiTY=0,9	PhiTF=0,75	
PhiS=0,9	PhiS-RI=1,	PhiST=0,9		
A=0,011	I33=1,126E-04	r33=0,103	S33=9,383E-04	Av3=0,007
J=1,040E-06	I22=3,923E-05	r22=0,061	S22=3,269E-04	Av2=0,002
E=210000000,	Fy=355000,	Ry=1,1	z33=0,001	Cw=0,
RLLF=1,	Fu=510000,		z22=4,980E-04	

STRESS CHECK FORCES & MOMENTS (Combo 1,2d+l+l,6w)

Location	Pu	Mu33	Mu22	Vu2	Vu3	Tu
0,	-70,769	-173,42	0,	-72,	0,	0,

PMM DEMAND/CAPACITY RATIO (H1-1b)
D/C Ratio: 0,529 = 0,014 + 0,515 + 0,
= (1/2)(Pr/Pc) + (Mr33/Mc33) + (Mr22/Mc22)

AXIAL FORCE & BIAXIAL MOMENT DESIGN (H1-1b)

Factor	L	K1	K2	B1	B2	Cm
Major Bending	4,	1,	1,587	1,	1,	1,
Minor Bending	1,	1,	1,	1,	1,	1,

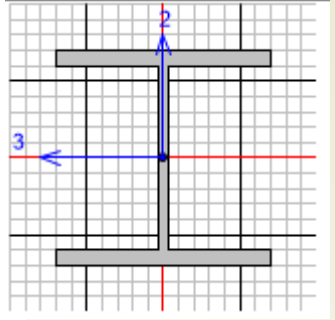
	Lltb	Kltb	Cb
LTB	1,	1,	1,199

	Pu Force	phi*Pnc Capacity	phi*Pnt Capacity
Axial	-70,769	2580,369	3386,7

	Mu Moment	phi*Mn Capacity	phi*Mn No LTB	phi*Mn Cb=1
Major Moment	-173,42	336,434	336,434	336,434
Minor Moment	0,	159,111		

SHEAR CHECK

			Vu Force	phi*Vn Capacity	Stress Ratio	Status Check
Major Shear	72,	511,2	0,141	OK		
Minor Shear	0,	1564,272	0,	OK		



Steel Stress Check Information (AISC 360-16)

Frame ID: 11
Design Code: AISC 360-16
Analysis Section: HE240B
Design Section: HE240B

COMBO ID	STATION LOC	MOMENT INTERACTION CHECK RATIO	=	AXL	+	B-MAJ	+	B-MIN	MAJ-SHR RATIO	MIN-SHR RATIO
1,2d+l+l,	0,	0,529(C)	=	0,014	+	0,515	+	0,	0,141	0,
1,2d+l+l,	0,5	0,422(C)	=	0,014	+	0,408	+	0,	0,141	0,
1,2d+l+l,	1,	0,315(C)	=	0,014	+	0,301	+	0,	0,141	0,
Nonlineer	0,	0,178(C)	=	0,023	+	0,155	+	0,	0,088	0,
Nonlineer	0,5	0,111(C)	=	0,023	+	0,088	+	0,	0,088	0,
Nonlineer	1,	0,043(C)	=	0,022	+	0,021	+	0,	0,088	0,

Modify/Show Overwrites
Overwrites

Display Details for Selected Item
Details

Strength Deflection

OK Cancel

Doubly and Singly Symmetric Members Subject to Flexure and Tension

For doubly symmetric members, C_b in Chapter F may be increased by $\sqrt{1+\frac{P_u}{P_{ey}}}$ for axial tension that acts concurrently with flexure,

where

$$P_{ey} = \frac{\pi^2 E I_y}{L_b^2}$$

UNSYMMETRIC AND OTHER MEMBERS SUBJECT TO FLEXURE AND AXIAL FORCE

$$\left|\frac{f_a}{F_a}+\frac{f_{bw}}{F_{bw}}+\frac{f_{bz}}{F_{bz}}\right| \le 1.0$$

For design according to Section B3.3 (LRFD)

f_a = required axial stress using *LRFD load combinations*, ksi (MPa)

F_a = $\phi_c F_{cr}$ = *design axial stress*, determined in accordance with Chapter E for compression or Section D2 for tension, ksi (MPa)

f_{bw}, f_{bz} = required flexural *stress* at the specific location in the cross section using LRFD load combinations, ksi (MPa)

F_{bw}, F_{bz} = $\dfrac{\phi_b M_n}{S}$ = *design flexural stress* determined in accordance with Chapter F, ksi (MPa). Use the section modulus for the specific location in the cross section and consider the sign of the stress.

ϕ_c = *resistance factor* for compression = 0.90

ϕ_t = resistance factor for tension (Section D2)

ϕ_b = resistance factor for flexure = 0.90